B1 sparse-guided passage selection: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Try to make the limited neural context see passages that a sparse statute signal judges useful. This changes evidence selection inside a single neural model rather than blending two final prediction streams.

Architecture and mechanism

A fit-only TF-IDF logistic salience model selects passage starts without dev or test labels. B1 encodes the selected windows; label-wise smooth-max pooling turns window evidence into statute probabilities.

Measured result

Scored submission	1016593
Codabench macro-F1	0.2417
Archived local metric	0.2464 official-dev macro-F1

The local official-dev macro-F1 is about 0.2464 and Codabench is 0.2417, below the plain LoRA baseline. The selected passages may have overemphasized lexical cues or disrupted broader context; this is a hypothesis, not a measured ablation.

Limitations and next check

The logged run does not isolate the passage selector from the changed pooling head, so causal attribution is limited. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase5_sparse_guided_model.py, phase5_sparse_guided_starts.py, phase5_multiwindow.py phase5_b1_train.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.